Artificial Intelligence (AI) & Prompt Engineering

Understanding the Fundamentals, Applications, and Future Scope

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Introduction to Artificial Intelligence (AI)

- • AI enables machines to perform tasks requiring human intelligence
- • Learning from data
- • Problem-solving
- • Decision-making
- • Speech recognition
- • Image understanding
- • Examples: ChatGPT, Siri, Alexa, Self-driving cars

Introduction to Prompt Engineering

- • Designing effective instructions for AI models
- • Improves output quality
- • Reduces ambiguity
- • Saves time and effort

- Example:
- Simple: Explain AI
- Better: Explain AI for a college student with examples

Importance and Applications of AI

- Importance:
- • Automation
- • Efficiency
- • Better decision-making

- Applications:
- • Healthcare
- • Education
- • Finance
- • Transportation
- • E-commerce

Importance and Applications of Prompt Engineering

- • Improves communication with AI
- • Produces relevant results
- • Enhances creativity

- Applications:
- • Content generation
- • Programming assistance
- • Research
- • Chatbots
- • Image generation

Key Concepts of AI

- • Machine Learning (ML)
- • Deep Learning
- • Natural Language Processing (NLP)
- • Computer Vision
- • Generative AI

Key Concepts of Prompt Engineering

- • Zero-shot Prompting
- • One-shot Prompting
- • Few-shot Prompting

- Best Practices:
- • Be specific
- • Provide context
- • Define output format
- • Use examples

Future Scope & Conclusion

- Future Scope:
- • Advanced automation
- • Smart healthcare
- • AI-powered education
- • Robotics
- • Scientific innovation

- Conclusion:
- AI and Prompt Engineering will shape the future of technology.

Zero-Shot, Few-Shot, and Chain-of-Thought Prompting

Prepared by: Nireeksha A P

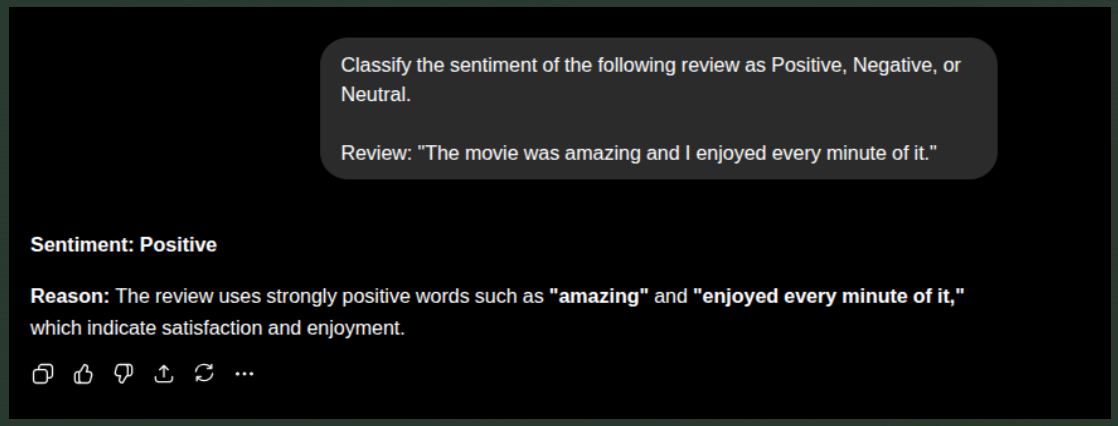
Advanced Prompt Engineering Techniques

Introduction to Prompt Engineering

- • Designing effective prompts for AI models
- • Improves accuracy and relevance
- • Enables complex reasoning and task completion
- • Used in education, coding, research, and business

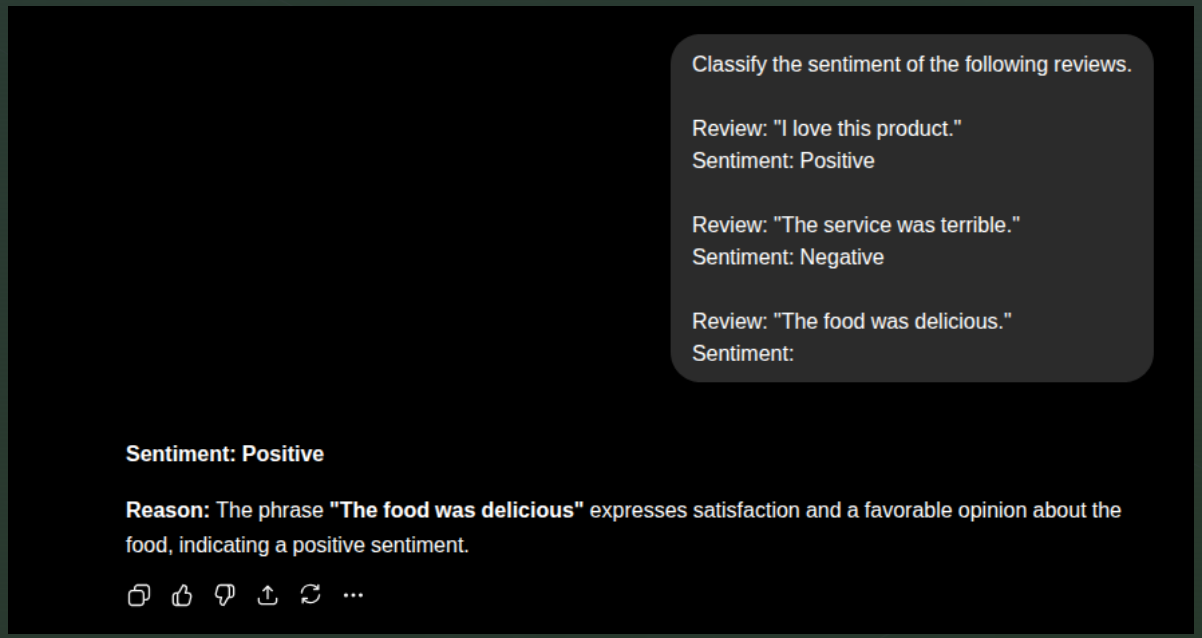
Zero-Shot Prompting

- Definition: Performing a task without examples.
- Example: Classify the sentiment of the following review as Positive, Negative, or Neutral.
- Review: "The movie was amazing and I enjoyed every minute of it."
- Applications: Translation, summarization, Q&A, sentiment analysis.
- Advantages: Simple and fast.
- Limitations: Less reliable for complex tasks.



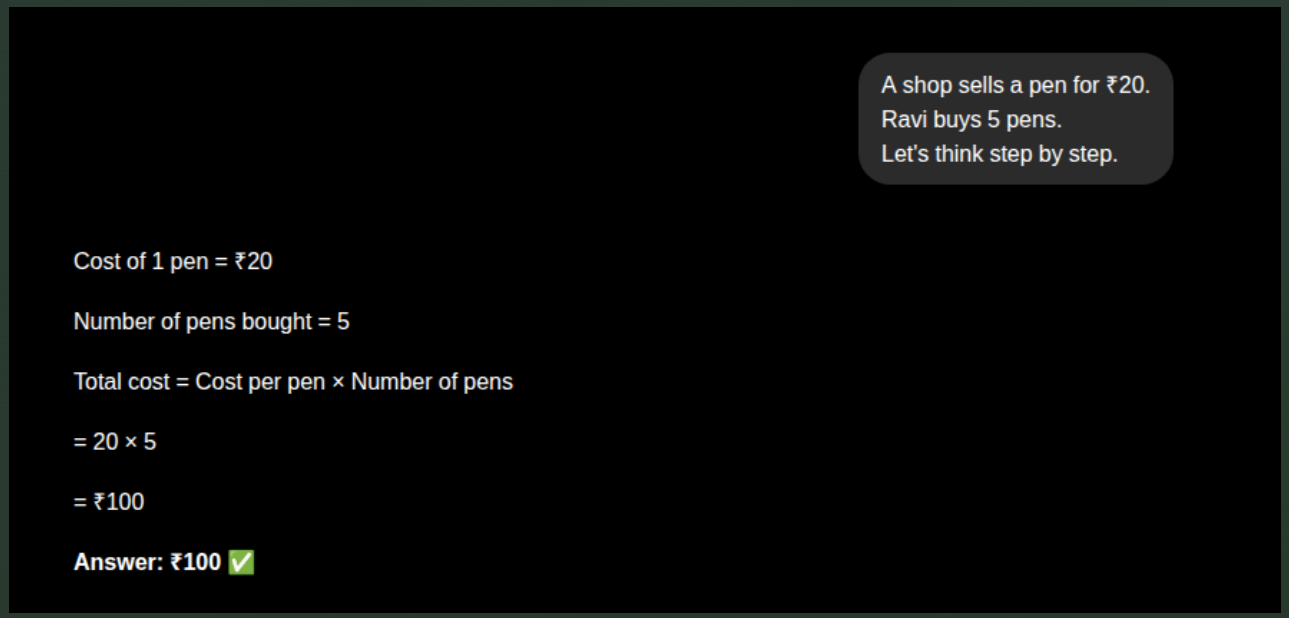
Few-Shot Prompting

- Definition: Providing a few examples before the task.
- Applications: Classification, chatbots, code generation, domain-specific tasks.
- Advantages: Better accuracy and consistency.
- Limitations: Requires carefully chosen examples.



Chain-of-Thought Prompting

- Definition: Encourages step-by-step reasoning.
- Applications: Math, logic, coding, decision-making.
- Advantages: Better reasoning and transparency.
- Limitations: Longer responses and possible reasoning errors.



Comparison of Techniques

- Zero-Shot: No examples, simple tasks.
- Few-Shot: Few examples, structured tasks.
- Chain-of-Thought: Reasoning-focused, complex problems.

Conclusion

- • Zero-shot is simple and fast
- • Few-shot improves performance with examples
- • Chain-of-thought enhances reasoning
- • Choose the technique based on task complexity